Day 1

# From Wound Swab to Genome

Genomic Surveillance of Surgical Site Infections & AMR
08:30–17:00

Course files: github.com/aryakaul/ughe-ssi-genomics-course

# Day 1 · Morning

| Time | Session |
|---|---|
| 08:30 | Welcome and knowledge check |
| 09:00 | **Lecture 1:** SSIs, AMR and why genomics? |
| 09:45 | **Lecture 2:** From sample to sequence |
| 10:30 | *Break* |
| 10:45 | **Lab 1:** Files, formats and your first Galaxy history |
| 12:15 | *Lunch* |

# Day 1 · Afternoon

| Time | Session |
|---|---|
| 13:00 | **Lecture 3:** How bacteria resist antibiotics |
| 13:45 | **Lab 2:** Quality control and genome assembly |
| 15:30 | *Break* |
| 15:45 | **Lab 3:** A look under the hood: the command line |
| 16:45 | Recap and exit ticket |

# Your course files

## github.com/aryakaul/ughe-ssi-genomics-course

- **Lab handouts:** the `labs/pdf/` folder
- **Slides:** the `slides/pdf/` folder
- **Everything at once:** green **Code** button → **Download ZIP**

# Before we start

- Log in to **usegalaxy.eu** and **pathogen.watch**
- Can’t log in? **Raise your hand now**: account emails can take a few minutes

**WORKING IN PAIRS**

**Driver:** hands on the keyboard · **Navigator:** reads the handout
You swap at each lab. Questions any time: **raise your hand**.

# Choose a code word

- Pick a **code word** you'll remember, e.g. a favourite food or place
- You'll use the **same code word** tomorrow, so we can compare before and after
- No names: the knowledge check is anonymous

# Knowledge check · Pre

- **Not graded**, and anonymous
- On your quiz sheet or a piece of paper, write your **code word** and **Pre**
- For each question, write its number and your answer (e.g. “1 b”)
- One question at a time on the screen

**KEY MESSAGE**

**It’s fine not to know an answer. Just leave it blank.**

KNOWLEDGE CHECK · QUESTION 1 OF 15

# Which question can whole-genome sequencing answer that standard culture + AST cannot?

a) Which antibiotic should this patient receive today

b) Whether isolates from several patients are the same strain

c) Whether the organism is Gram-positive

d) Whether the patient has a fever

KNOWLEDGE CHECK · QUESTION 2 OF 15

# Do you need to own a sequencer to do genomic surveillance? (Yes / No) Briefly explain why.

Write your answer on your paper

KNOWLEDGE CHECK · QUESTION 3 OF 15

# Name two bacterial species that commonly cause surgical site infections.

Write your answer on your paper

KNOWLEDGE CHECK · QUESTION 4 OF 15

# An Illumina sequencing run typically produces reads that are:

a) 100–300 bases long and highly accurate

b) 10,000+ bases long

c) the whole genome in one read

d) protein sequences

KNOWLEDGE CHECK · QUESTION 5 OF 15

# Which technology is better at showing whether a resistance gene is on a plasmid?

a) Illumina short reads alone

b) Oxford Nanopore long reads

c) AST

d) MLST

KNOWLEDGE CHECK · QUESTION 6 OF 15

# A genome is 5 million bases. You have 1 million reads of 150 bases. What is the approximate coverage?

a) 3×

b) 30×

c) 150×

d) 300×

# What does a FASTQ file contain that a FASTA file does not?

a) Sequence names

b) DNA letters

c) Quality scores

d) Gene names

KNOWLEDGE CHECK · QUESTION 8 OF 15

# After assembly, which result usually means a *better* assembly?

a) More contigs and a smaller N50

b) Fewer contigs and a larger N50

c) A total length twice the expected genome size

d) No contigs

# The gene *mecA* in *S. aureus* indicates:

a) Vancomycin resistance

b) MRSA (methicillin resistance)

c) Carbapenem resistance

d) A virulence toxin

KNOWLEDGE CHECK · QUESTION 10 OF 15

# *blaNDM-1* is an example of:

a) An ESBL

b) A carbapenemase

c) An efflux pump

d) A housekeeping gene

KNOWLEDGE CHECK · QUESTION 11 OF 15

# The genome shows no known resistance gene, but AST says “resistant”. Give one possible explanation.

Write your answer on your paper

# Two *Klebsiella* isolates are ST15 and ST307. Could they be part of the same recent transmission chain?

a) Yes, very likely

b) No, they are different strain types

c) Only if they are both ESBL

KNOWLEDGE CHECK · QUESTION 13 OF 15

# On a phylogenetic tree, what tells you how genetically different two isolates are?

a) Their order from top to bottom

b) The branch lengths between them

c) The colour of the labels

KNOWLEDGE CHECK · QUESTION 14 OF 15

# Four isolates are 2–6 SNPs apart and the patients overlapped on the same ward. This most strongly suggests:

a) Unrelated infections

b) Possible transmission / outbreak

c) Contamination of the sequencer

KNOWLEDGE CHECK · QUESTION 15 OF 15

# List three pieces of metadata that must be recorded with every isolate for surveillance.

Write your answer on your paper

# How confident are you?

Rate each from **1** (not at all) to **5** (very confident):

- I can explain what a genome sequence can tell us about an SSI
- I can run a tool in Galaxy
- I can identify resistance genes from a genome
- I can interpret a phylogenetic tree
- I can plan a genomic surveillance project

# N1 · About you

**Your main role:**
Clinician · Nurse / midwife · Laboratory · Public health / IPC · Research · Other

**Your institution:**
UGHE · Butaro Hospital · Other

# N2 · At your site, is there…

Answer **Yes**, **No** or **Not sure**:

1. culture and antibiotic susceptibility testing?
2. long-term storage of bacterial isolates (e.g. a freezer)?
3. internet reliable enough to use web tools like Galaxy?

# N3

## What is the biggest barrier to using genomic data in your work?

Lecture 1

# Surgical Site Infections, AMR and Why Genomics?

Day 1 · 09:00–09:45 · 45 min
≈ 30 min talk + 15 min discussion

LO1: explain how WGS strengthens SSI and AMR surveillance compared with culture and AST alone

# Think of the last surgical patient whose infection didn’t respond to first-line antibiotics.

What did we learn about where it came from?

# What is a surgical site infection?

- An infection at or near the incision
- Within **30 days** of surgery (**90 days** with an implant)
- Three levels:
  1. Superficial incisional
  2. Deep incisional
  3. Organ / space

# The burden

- The **most common healthcare-associated infection** in low- and middle-income countries (WHO)
- Post-caesarean SSI is a major problem in sub-Saharan Africa, including rural Rwanda
- Consequences:
  - longer hospital stays and reoperation
  - catastrophic costs for households
  - death

# SSI after caesarean section in Rwanda

**10.9%**
Kirehe District Hospital
Nkurunziza et al. 2019

**5.7%**
Kirehe, later cohort
Velin et al. 2021

**6.85%**
National pooled estimate
Sibomana et al. 2024

# Resistance in SSI isolates from Kirehe

**68%**
of isolates were Gram-negative

**0%**
of Gram-negatives were susceptible to ampicillin

**92%**
were intermediate or resistant to **ceftriaxone**

Velin et al., *Ann Glob Health* 2021. Ceftriaxone is the standard empiric drug.

# The gap

- Culture and AST capacity is still limited (Umutesi et al. 2021)
- We **don’t know which resistance genes or clones** are behind these numbers

**KEY MESSAGE**

**That is the gap genomics fills.**

**DISCUSS**

What fraction of SSIs in our hospitals get a culture? An AST?

# AMR: the global picture (2019)

**1.27 M**
deaths **directly caused**
by bacterial AMR

**4.95 M**
deaths **associated**
with bacterial AMR

- The highest death rate was in **sub-Saharan Africa**
- 5 of the 6 leading pathogens are common causes of SSI

GRAM study, *Lancet* 2022

# SSI pathogens to know

| Pathogen | Resistance to watch |
|---|---|
| *Staphylococcus aureus* | MRSA (*mecA* / *mecC*) |
| *Escherichia coli* | ESBLs (CTX-M), carbapenemases |
| *Klebsiella pneumoniae* | ESBLs, carbapenemases (KPC, NDM, OXA-48) |
| *Acinetobacter baumannii* | Carbapenemases (OXA-23) |
| *Pseudomonas aeruginosa* | Efflux, porin loss, VIM / IMP |

**ESKAPE:** *Enterococcus*, *S. aureus*, *Klebsiella*, *Acinetobacter*, *Pseudomonas*, *Enterobacter*

# What our current tools tell us

**CULTURE AND AST ANSWER**

- **Which organism?**
- **Which drugs work?**

Essential for treating **this patient**.

**THEY CAN'T ANSWER**

- Are these 5 patients infected with the **same strain**?
- Is resistance spreading on a **plasmid** between species?
- Is a known **high-risk clone** circulating?

# What a genome adds

A bacterial genome is 2–6 million letters (A, C, G, T). From one genome we get:

**1 · Species**
confirms or corrects the lab ID

**2 · Strain type**
MLST: “the bacterium’s surname”

**3 · AMR genes**
and **where** they sit: chromosome or plasmid

**4 · Virulence genes**
toxins and other factors

**5 · Relatedness**
SNP distance: the basis of outbreak detection

# Case example: an outbreak response

MRSA on a **neonatal unit** in Cambridge, UK

- **Köser et al., *NEJM* 2012:** separated the linked outbreak isolates from unrelated MRSA that looked identical on routine lab tests
- **Harris et al., *Lancet Infect Dis* 2013:** traced transmission into the community and found a **staff carrier**, which ended the outbreak

**KEY MESSAGE**

**Genomics changed the infection-control decision. We'll use this reasoning in Lab 6.**

# Genomic surveillance ≠ owning a sequencer




- A reference lab or service provider can do the **sequencing**
- The **expertise stays local**: choosing isolates, recording metadata, analysing, interpreting, communicating
- Africa CDC and national labs are building this network

# The next two days

**DAY 1 · THE DATA**

What sequencing produces, and how to turn it into a genome

**DAY 2 · THE ANSWERS**

Resistance, strain types, outbreaks, and a sustainable programme

# Discussion

15 min · pairs, then the whole room

1. Describe one situation in your work where knowing **“are these infections linked?”** would have changed what you did.
2. What happens to bacterial isolates in our labs after AST is done? **Are they stored?**

Lecture 2

# From Sample to Sequence

Day 1 · 09:45–10:30 · 45 min

LO2: describe Illumina and Nanopore outputs and choose between them · LO6 (part): understand the outsourcing workflow

# The journey of an isolate

**Wound swab** → **Culture** → **Pure isolate**

**DNA extraction** → **Library prep** → **Sequencing** → **Data files** → **Analysis**

KEY MESSAGE

**Everything up to the pure isolate (dark boxes) already happens in our labs.**

# We sequence isolates, not swabs

- An **isolate** is a pure culture of one bacterium, so its genome is clean
- Sequencing a swab directly (**metagenomics**) mixes human DNA with many microbes: harder and more expensive
- For SSI surveillance, **isolate sequencing is the standard**

**KEY MESSAGE**

**Store your isolates (–80 °C or glycerol stocks). You can’t sequence what you threw away.**

# DNA extraction and library preparation

- **Extraction:** break open the cells and purify the DNA
- **Library prep:** cut or tag the DNA and add short “adapter” sequences so the machine can read it
- **Multiplexing:** many samples are barcoded and run together, which lowers the cost per genome

> You don’t need to do this yourself. But poor-quality DNA gives poor-quality data (Lab 2).

# Two main technologies

| | Illumina (short reads) | Oxford Nanopore (long reads) |
|---|---|---|
| **Read length** | 100–300 letters, in pairs | thousands to 100,000+ letters |
| **Accuracy** | very high | now high with modern chemistry |
| **Best for** | SNP-level outbreak work | complete genomes, **plasmids** |
| **Machine** | large, central labs | MinION is pocket-sized |
| **Files** | FASTQ pair (R1 + R2) | one FASTQ |

Jigsaw analogy: Illumina gives many tiny pieces; Nanopore gives fewer, bigger pieces.

# What is a read?

- A **read** is one piece of DNA sequence produced by the machine
- One bacterial genome gives **hundreds of thousands to millions** of reads
- We put the reads back together into the genome

> Like rebuilding a shredded book from many overlapping copies.

# Coverage (depth)

**Coverage** = how many reads, on average, cover each position

$$(\text{number of reads} \times \text{read length}) \div \text{genome size}$$

$$1{,}000{,}000 \times 150 \div 5{,}000{,}000 = \mathbf{30\times}$$

**KEY MESSAGE**

**Aim for ≥ 30× with Illumina and ≥ 40–50× with Nanopore.**

# Which should I use?

| Question | Best choice |
|---|---|
| Is this an outbreak? (fine SNP differences) | Illumina |
| Is resistance on a plasmid moving between species? | Nanopore or hybrid |
| Need results fast, locally | Nanopore |
| Lowest cost per genome at large scale | Illumina (batched) |
| Best of both | **Hybrid** assembly |

**KEY MESSAGE**

**Your sequencing partner may use either. This course teaches both.**

# Outsourcing: how it works

**Agree on a partner** → **Ship isolates or DNA** → **Partner sequences** → **You analyse**

- **Partners:** national reference lab, Africa CDC network, academic collaborator, or a commercial service
- **Shipping:** follow their biosafety rules; you may need permits and a material transfer agreement
- **You get back:** FASTQ files, sometimes assemblies and reports. **Keep the data.**

# Questions to ask a sequencing partner

**Platform?**
Illumina, Nanopore, or both

**Coverage?**
≥ 30× Illumina

**Raw reads?**
Make sure FASTQ files are included

**Turnaround?**
Weeks from shipping to data

**Cost?**
Per isolate, including shipping

**Data sharing?**
Who owns and publishes the data

# Metadata: the most underrated part

A genome without **when, where and from whom** is almost useless for surveillance.

**Minimum fields**

- isolate ID · collection date
- facility · ward · specimen type
- surgery type and date
- AST results

**PROTECT PRIVACY**

**Never** put names or patient IDs in shared files. Use coded study IDs.

Standard: **PHA4GE** contextual data specification

# What files come back?

| File | What it is |
|---|---|
| `sample_R1.fastq.gz` + `sample_R2.fastq.gz` | Illumina reads (a pair) |
| `sample.fastq.gz` | Nanopore reads |
| `sample.fasta` | Assembly (sometimes) |
| QC report | Sometimes |

> In Lab 1 we'll open these files and look inside.

# Quick check (show of hands)

1. You want to know if **6 MRSA isolates from one ward** are an outbreak. Which data would you request?

2. You suspect a **carbapenemase gene is jumping** between *Klebsiella* and *E. coli*. Which data would you request?

# Quick check: answers

1. **Illumina, ≥ 30×**, with collection dates and ward metadata

2. **Nanopore or hybrid**, to resolve the plasmids

# Break

Back at 10:45 for Lab 1

GENOMIC SURVEILLANCE OF SURGICAL SITE INFECTIONS & AMR · UGHE–HMS

Lab 1

# Files, Formats and Your First Galaxy History

Day 1 · 10:45–12:15 · 90 min
Pairs: driver + navigator · swap at Part C

LO2, LO3: recognise genomic file formats, read a quality score, and run a tool in Galaxy

# Open your handout

## Lab 1 handout

`Lab1_files_formats_galaxy.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

# Plan for this lab

| Part | What | Time |
|---|---|---|
| A | Files and folders | 15 min |
| B | Looking inside the formats | 25 min |
| C | Your first Galaxy history | 40 min |
| | Wrap-up | 10 min |

# Before you start

- Log in to `usegalaxy.eu`
- **Driver:** hands on the keyboard
- **Navigator:** reads the handout and spots mistakes

> Stuck or have a question? **Raise your hand** and the instructor will come to you.

# What's in a file name?

`isolate01_R1.fastq.gz`

| `isolate01` | `R1` | `.fastq` | `.gz` |
|---|---|---|---|
| sample name | read 1 of a pair | format | compressed |

**CHECK YOURSELF**

What do you expect is inside `UGHE-SSI-0042_R2.fastq.gz`?

# Two more things to know

- **Plain-text files** (FASTA, FASTQ, CSV) are just letters: any text editor can open them
- **Genomic files are big**: 200 MB–1 GB per isolate

KEY MESSAGE

**That’s why we analyse in Galaxy, on computers on the internet, not on our laptops.**

# FASTA: just sequence

```
>contig_1 length=120
TTCGCTCTATTGACTACGACGCGCTCATTCCCTTGTCGGAGAGTTATGGAACAAGGACGC
TGTCTGAGACTAGAAGACAGATAGTGCACACGACCGGCGTCGGAGAAACTCTATTTGCCG
>contig_2 length=60
CCTGACAAGTCAATGCGATCCGTAGGGGCAGCGCAGTATGCCAAGACTATAGGCACTGTC
```

- Each entry starts with > and a **name**; the sequence follows
- **Assemblies** (genomes) are stored in FASTA

**CHECK YOURSELF**

How many sequences? How many bases in total?

# FASTQ: sequence + quality (raw reads)

Every read takes **exactly 4 lines**:

| Line | Example | Meaning |
|---|---|---|
| 1 | `@SSI_isolate_01_read1` | @ + read name |
| 2 | `GCTAAAGACAATTACATAACATACACGTCAGCACGAAACT` | the DNA sequence |
| 3 | `+` | separator |
| 4 | `GHHGFGHIIIIIFIGGHGIFIIGHIHIGHIIGFHHDACBC` | quality, one symbol per base |

# Reading a quality score

| Symbol | # | + | 5 | ? | I |
|---|---|---|---|---|---|
| **Q score** | 2 | 10 | 20 | 30 | 40 |
| **Chance it's wrong** | ≈ 63% | 1 in 10 | 1 in 100 | 1 in 1,000 | 1 in 10,000 |

**RULE OF THUMB**

Capital letters (A–J) and ? or @ are good (Q30 and above).
Symbols like #, $, %, & are bad (below Q10).

# Check yourself: read 4

```
IIHFGIIIFIIIIIFIIHFIIIIHI#'&'##&%''''$(%
```

**CHECK YOURSELF**

1. Which part of read 4 (start or end) has poor quality?
2. What should we do with those bases?

# CSV / TSV: tables

```
isolate_id,organism,collection_date,ward
UGHE-SSI-0001,Klebsiella pneumoniae,2026-03-02,Maternity
UGHE-SSI-0002,Escherichia coli,2026-03-05,Surgical
```

- One row per isolate, one column per piece of information
- Commas (CSV) or tabs (TSV) separate the columns
- They open in Excel or Google Sheets

# Swap driver and navigator

The navigator takes the keyboard; the driver reads the handout

# A tour of Galaxy

- **Left, Tools:** search box and tool list
- **Centre:** tool forms and results
- **Right, History:** your files (**datasets**), in order

| Colour | Meaning |
|---|---|
| green | finished |
| yellow | running |
| grey | queued |
| red | error |

# Create a history and upload data

1. History panel → **“+”** → rename it `Lab1 – formats`
2. **Upload** → **Paste/Fetch data** → paste the **4 reads** from your handout
3. **Type:** `fastqsanger` · **Name:** `tiny_reads.fastq`
4. **Start** → **Close**
5. When it turns **green**: click the name, then the **eye icon**

PART C4

# Run your first tool: count lines

1. Search the tools: **Line/Word/Character count**
2. “Text file” → `tiny_reads.fastq`
3. **Run tool**, then view the result (eye icon)

> **CHECK YOURSELF**
>
> How many lines? How many reads is that? *(4 lines per read)*

PART C5

# Run a real QC tool: FastQC

1. Search the tools: **FastQC** → input `tiny_reads.fastq` → **Run tool**
2. Open the **Webpage** output (eye icon)
3. Find **“Per base sequence quality”**

**CHECK YOURSELF**

- Does quality go up or down towards the end of the reads?
- Would you trust a report based on only 4 reads?

PART C6

# Your history is a lab notebook

- Click a dataset, then the **“i”** (Dataset details) icon
- Galaxy records the **exact** tool, version and settings

> **KEY MESSAGE**
>
> **This is reproducibility: anyone can see and repeat what you did.**

# Wrap-up

| Format | Holds |
|---|---|
| **FASTA** | sequences (assemblies) |
| **FASTQ** | reads + quality (raw data) |
| **CSV / TSV** | tables (metadata, results) |

**Upload** → **Choose tool** → **Run** → **View**

After lunch: **real** reads from a bacterial isolate.

Adapted from the Galaxy Training Network “Introduction to Galaxy” and “Quality Control” tutorials (CC-BY 4.0).

# Lunch

Back at 13:00 for Lecture 3

Lecture 3

# How Bacteria Resist Antibiotics

Day 1 · 13:00–13:45 · 45 min

LO4 (part): understand the genetic basis of AMR so that genome results can be interpreted

# Four ways to survive an antibiotic

**1 · Destroy the drug**
Enzymes such as β-lactamases cut penicillins and cephalosporins

**2 · Change the target**
The drug no longer binds (PBP2a in MRSA; *gyrA* mutations)

**3 · Pump it out**
Efflux pumps

**4 · Keep it out**
Fewer or changed porins in Gram-negatives

# Where resistance comes from: two routes

**ACQUIRED GENE**

The bacterium picks up a **new gene** from another bacterium

*mecA*, *blaCTX-M-15*, *blaNDM-1*

**Genome:** is the gene present or absent?

**CHROMOSOMAL MUTATION**

A change in the bacterium's **own gene**

*gyrA* S83L (ciprofloxacin)

**Genome:** compare with a reference and look for specific mutations

# Plasmids: why resistance spreads so fast

- **Plasmids:** small circles of DNA that move **between bacteria, even between species**
- **Transposons and integrons:** “cut-and-paste” elements that move genes around
- One plasmid can carry **many** resistance genes at once

**KEY MESSAGE**

**A resistant plasmid can spread even when the strain is not spreading. Genomics can tell these apart.**

# The big four to recognise

***mecA***
MRSA: resistant to almost all β-lactams

***blaCTX-M* (ESBL)**
Resistant to 3rd-generation cephalosporins such as ceftriaxone

**Carbapenemases**
KPC, NDM, OXA-48, VIM, IMP, OXA-23: resistant to carbapenems, the “last line”

***gyrA* / *parC* mutations**
Fluoroquinolone (ciprofloxacin) resistance

You don’t need to memorise gene names. The tools report them, and the glossary has a cheat sheet.

# Other mechanisms you'll see

| Genes | Resistance to |
|---|---|
| *blaSHV*, *blaTEM* (ESBL variants) | 3rd-generation cephalosporins |
| *aac*, *aph*, *ant*, *armA*, *rmtB* | Gentamicin, amikacin |
| *sul1*, *sul2*, *dfrA* | Co-trimoxazole |
| *tet(A)*, *tet(M)* | Tetracyclines |

# Reading gene names

## blaCTX-M-15

| bla | CTX-M | 15 |
|---|---|---|
| β-lactamase | family | variant |

- aac(6')-Ib-cr: aminoglycoside enzyme; “cr” = also affects ciprofloxacin
- mecA: methicillin resistance

**1-MINUTE EXERCISE**

What do you predict blaNDM-1 does?

# Can the genome predict the AST result?

**WORKS VERY WELL**

- *S. aureus*
- β-lactams in *E. coli* and *Klebsiella*

**WORKS LESS WELL**

- Resistance depends on how much a gene is **expressed**
- **Combinations** of weak mechanisms
- **Unknown** mechanisms not yet in databases

**KEY MESSAGE**

**The genome complements AST. It does not replace AST for treating patients.**

# When the genome and the AST disagree

| Genome says | AST says | Possible explanations |
|---|---|---|
| Resistance gene | Susceptible | Gene not expressed or broken · lab error · mixed culture |
| No gene found | Resistant | Unknown mechanism · mutation not in database · low coverage · mixed culture |

> Disagreements are **useful**: they point to new mechanisms or lab QC problems. We'll practise this in Lab 4.

# AMR tools and databases (preview of Lab 4)

| Tool | From | Notes |
|---|---|---|
| **AMRFinderPlus** | NCBI, USA | Genes + mutations; organism-aware |
| **ResFinder / PointFinder** | CGE, Denmark | Web tool; predicts phenotype |
| **CARD / RGI** | McMaster, Canada | Very comprehensive |
| **ABRicate** | Community | Fast screening, several databases |

> Tools may disagree slightly. Always report which tool and database version you used.

# Summary

- Resistance = **genes** (often mobile) + **mutations**
- Plasmids let resistance spread **between species**
- Genome predictions are powerful but have limits: **always read them next to AST**

> Next: let’s make some genomes (Lab 2).

Lab 2

# Quality Control and Genome Assembly

Day 1 · 13:45–15:30 · 105 min
Pairs · half the room on the Illumina track, half on the Nanopore track

LO3: perform QC and assembly on raw reads in Galaxy, and judge whether an assembly is good enough to use

# Open your handout

## Lab 2 handout

`Lab2_qc_assembly.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

# One isolate, sequenced twice

- An **MRSA** strain (KUN1163), sequenced on **Illumina** and on **Oxford Nanopore**
- Half the pairs take each track; at the end you compare

**STRATEGY**

Assembly takes **20–60 min**. Start it early and interpret QC while it runs.
Not finished by **15:00**? Use the fallback (Part E).

# Plan for this lab

| Part | What | Time |
|---|---|---|
| A | Set up and upload | 10 min |
| B–D | Check quality → clean → start assembly | 35 min |
| F | Interpret QC while it runs | 20 min |
| G | Assess the assembly | 15 min |
| H | Compare tracks | 10 min |

# Upload your track's reads

1. New history: `Lab2 – Illumina` **or** `Lab2 – Nanopore`
2. **Upload** → **Paste/Fetch data** → paste the URL(s) **from your handout**
3. **Type:** `fastqsanger.bz2` → **Start**
4. Rename: `illumina_R1` + `illumina_R2`, **or** `nanopore_reads`

> **CHECK YOURSELF**
>
> Why are there two Illumina files but only one Nanopore file?

# Illumina: check, clean, assemble

**FastQC** → **fastp** → **Shovill**

1. **FastQC** on `illumina_R1`, then on `illumina_R2`
2. **fastp:** Paired · R1 and R2 · defaults
3. **Shovill:** Paired-end · the two fastp outputs · defaults

> Shovill can’t see the fastp files? Pencil icon → **Datatypes** → `fastqsanger.gz`

# Nanopore: check, filter, assemble

**NanoPlot** → **filtlong** → **Flye**

1. **NanoPlot** on `nanopore_reads`
2. **filtlong:** Min. length `1000` · defaults
3. **Flye:** filtlong output · Mode **Nanopore corrected** · defaults

# Assembly running?

Don’t wait for it: go on to Part F and interpret your QC

# Interpret your QC

| | Your track |
|---|---|
| Number of reads | |
| Typical read length | |
| Total bases (≈ reads × length) | |
| **Coverage** = total bases ÷ 2.8 Mb | |
| Quality good enough? Why? | |

# Check yourself: quality

**CHECK YOURSELF**

1. Is your coverage above the minimum?
   (≈ 30× Illumina, ≈ 40–50× Nanopore)
2. Look at the report from a **bad** run on the screen. What looks different?

# Assess the assembly with QUAST

**Quast** → your Shovill `contigs` or Flye `assembly` → defaults → HTML report

| Metric | Good sign for *S. aureus* |
|---|---|
| **# contigs** (pieces) | Illumina: tens to ≈ 100 · Nanopore: 1–5 |
| **Total length** | ≈ 2.7–3.0 Mb |
| **N50** | Bigger is better |
| **GC (%)** | ≈ 33% |

# Check yourself: assembly

**CHECK YOURSELF**

1. Is the total length close to the expected genome size?
2. What would a total length of **5.6 Mb** suggest?
3. Would you accept this assembly for AMR analysis? Why?

# Compare the two tracks

| | Illumina | Nanopore |
|---|---|---|
| # contigs | | |
| Largest contig | | |
| Total length | | |
| N50 | | |

- Which is more **complete**?
- Which is more **accurate**?
- Gene on a **plasmid**? Which would you prefer?
- What would a **hybrid** give you?

# Still running at 15:00?

1. Open the shared history link (handout Part E):
   **usegalaxy.eu/u/aryakaul/h/ughe-ssi-course-precomputed**
2. Click **Import**
3. Use the Shovill or Flye assembly and its QUAST report
4. Continue from Part G

> Your own job **keeps running**. Check it tonight!

# Summary

**Illumina** → many accurate contigs

**FastQC** → **fastp** → **Shovill** → **QUAST**

**Nanopore** → few long contigs (≈ complete genome)

**NanoPlot** → **filtlong** → **Flye** → **QUAST**

Tomorrow: **what is in this genome?**

Adapted from the Galaxy Training Network “Bacterial Genome Assembly” tutorials (Shovill, Flye) (CC-BY 4.0).

# Break

Back at 15:45. First thing: start JupyterLab (Lab 3, Part A)

Lab 3

# A Look Under the Hood: the Command Line

Day 1 · 15:45–16:45 · 60 min
Pairs · swap driver and navigator at Part C

LO3: understand what Galaxy does behind the buttons, and take a first step towards independent analysis

# Open your handout

## Lab 3 handout

`Lab3_command_line_taster.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

# Why bother?

- Every Galaxy tool you clicked today is a **command-line program**
- A dozen commands let you look inside huge files and follow online tutorials

**KEY MESSAGE**

**You don’t need to become a programmer. This is the most durable skill in the course.**

# Golden rules

- Commands are **case-sensitive**, and spaces matter
- **Enter** runs a command · ↑ repeats the last one
- **Tab** completes file names
- **Ctrl + C** stops a running command

# Open a terminal inside Galaxy

1. Run the tool **Interactive JupyterLab Notebook** (no changes)
2. After 1–3 min: **User** → **Active InteractiveTools** → click the link
3. In JupyterLab: **File** → **New** → **Terminal**
4. You'll see a prompt ending in `$`. Type here.

> Not working? Use your laptop's **Terminal** (macOS) or **Git Bash** (Windows).

# Where am I? Moving around

| Command | What it does |
|---|---|
| `pwd` | **p**rint **w**orking **d**irectory: where am I? |
| `ls` | list the files here |
| `mkdir ssi_course` | make a new folder |
| `cd ssi_course` | go into a folder |
| `cd ..` | go up one level |

PART B

# Try it

Type each line, then press Enter:

```
pwd
ls
mkdir ssi_course
cd ssi_course
cd ..
pwd
cd ssi_course
```

**CHECK YOURSELF**

Draw a little “tree” of the folders you moved through.

# Swap driver and navigator

The navigator takes the keyboard; the driver reads the handout

# Download the reads

The same Illumina read 1 file you used in Lab 2:

```
wget https://zenodo.org/…/DRR187559_1.fastqsanger.bz2
ls -lh
```

- Copy the **full URL** from your handout (Part C1)
- `ls -lh` lists files with their sizes (K, M, G)

PART C2

# Peek inside a compressed file

```
bzcat DRR187559_1.fastqsanger.bz2 | head -8
```

- `bzcat` unzips the file as it reads it
- `head -8` shows the first 8 lines
- `|` is a **pipe**: it sends one command's output into the next

**CHECK YOURSELF**

How many reads did you see? Point to the first read's quality line.

PART C3

# Count the reads

FASTQ has 4 lines per read: **count lines, then divide by 4**

```
bzcat DRR187559_1.fastqsanger.bz2 | wc -l
```

**CHECK YOURSELF**

How many reads? Does it match “Total sequences” in your Lab 2 FastQC report?

PART C3

# A trap

Someone suggests counting reads like this:

```
bzcat DRR187559_1.fastqsanger.bz2 | grep -c "^@"
```

It counts lines that start with `@`.

> **CHECK YOURSELF**
>
> Why could this give the **wrong** answer?
> *Hint: which quality score is @?*

PART C4

# Search inside the file

How many reads contain a run of 10 A's?

```
bzcat DRR187559_1.fastqsanger.bz2 | grep -c "AAAAAAAAAA"
```

`grep` finds lines containing a pattern; `-c` counts them

# From reads to genomes

Download the assembly (URL in your handout, Part D), then:

```
head -3 DRR187559_contigs.fasta
grep -c ">" DRR187559_contigs.fasta
```

**CHECK YOURSELF**

What does `grep -c ">"` count? Compare it with **# contigs** in QUAST.

# Galaxy ↔ command line

| In Galaxy… | …on the command line |
|---|---|
| Upload → Paste URL | `wget URL` |
| Eye icon | `head`, `less` |
| Line/Word/Character count | `wc -l` |
| History | `history` (try it!) |

> When you finish: **User** → **Active InteractiveTools** → **Stop** JupyterLab.

# Cheat sheet 1: moving around

| | |
|---|---|
| `pwd` | where am I? |
| `ls` · `ls -lh` | list files · with sizes |
| `cd folder` · `cd ..` | go into a folder · go up |
| `mkdir name` | make a folder |
| `wget URL` | download a file |
| `history` | commands you have run |

# Cheat sheet 2: looking inside files

| | |
|---|---|
| `head -n 8 file` | first 8 lines |
| `wc -l file` | count lines |
| `grep "text" file` | find lines containing text (`-c` counts them) |
| `bzcat` · `zcat` | read a .bz2 · .gz file without unzipping |
| `cmd1 \| cmd2` | pipe: send cmd1's output into cmd2 |

**Want more?** Software Carpentry **Unix Shell** lesson: `swcarpentry.github.io/shell-novice`

Adapted from The Carpentries "The Unix Shell" and the GTN "CLI basics" tutorial (CC-BY 4.0).

# Today we…

1. Saw why genomics matters for SSIs and AMR
2. Learned what sequencing produces: reads, FASTQ, coverage
3. Ran our first tools in Galaxy
4. Turned raw reads into a genome and checked its quality
5. Looked under the hood with the command line

# Exit ticket

## What is still unclear?

One question on a slip of paper. No names.

> I’ll answer them first thing tomorrow.

# See you tomorrow at 08:30

Day 2: from genomes to surveillance action